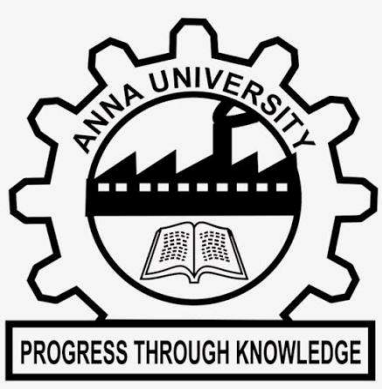
SENTIMENT ANALYSIS OF CUSTOMER REVIEWS USING NATURAL LANGUAGE PROCESSING AND MACHINE LEARNING

A SUMMER INTERNSHIP REPORT

Submitted by

L.VIGNESHWARAN (810023205062)



DEPARTMENT OF INFORMATION TECHNOLOGY

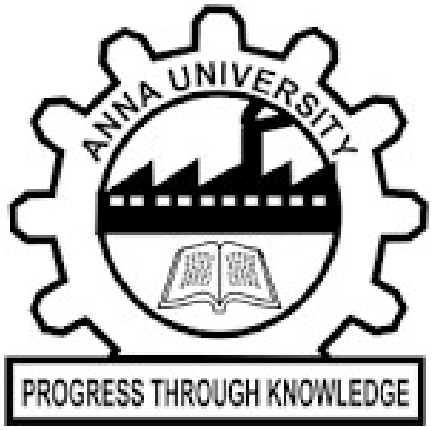
UNIVERSITY COLLEGE OF ENGINEERING (BIT CAMPUS),

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report **"Sentiment Analysis of Customer Reviews Using Natural Language Processing and Machine Learning"** is the Bonafide work of **L. VIGNESHWARAN (810023205062)** and has satisfactorily completed the course of **IT3711 - SUMMER INTERNSHIP** for **SEVENTH** semester, **B.Tech. INFORMATION TECHNOLOGY** during the academic year 2026-2027**.**

Faculty in charge **Head of the department**

Submitted for the practical examination held on

ABSTRACT

This internship report presents the knowledge and practical experience gained during my 30-day internship at **Fantasy Solution Technology** in the domain of **Data Science and Machine Learning**. The internship provided an opportunity to apply theoretical concepts to real-world data analysis and machine learning applications while improving my technical, analytical, and problem-solving skills. During the internship, I studied and applied various stages of the data science lifecycle, including **data collection, data preprocessing, exploratory data analysis, data visualization, feature extraction, machine learning model development, and model evaluation**. As part of the internship, I developed a **Sentiment Analysis System for Customer Reviews**, which analyses textual customer reviews and classifies them into **positive, negative, and neutral sentiments** using Natural Language Processing (NLP) and machine learning techniques. The project involved preprocessing textual data, extracting meaningful features using **TF-IDF**, training and comparing different machine learning algorithms, and evaluating their performance using metrics such as **accuracy, precision, recall, and F1-score**. Through this internship, I gained practical experience in **Python, Pandas, NumPy, Matplotlib, Seaborn, NLTK, Scikit-learn, Natural Language Processing, and Machine Learning**. The internship strengthened my understanding of data-driven problem solving and enhanced my confidence in developing practical machine learning solutions for real-world applications.

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CHAPTER 1: INTRODUCTION & OBJECTIVES

1.1 COMPANY PROFILE



FANTASY SOLUTION

Fantasy solution as a leading IT solution and service provider, provides innovative information technology - enabled solutions and services to meet the demands arising from social transformation, shaping new lifestyles for individuals and creating values for the society.

Focusing on software technology, Fantasy solution provides industry solutions and product engineering solutions, related software products & platforms, and services, through seamless integration of software and services, software and manufacturing, as well as technology and industrial management capacity.

Fantasy solution helps industry customers establish best practices in business development and management. The fantasy solution serves include real time projects, web designing, web hosting, software development and training etc, in many of which, has a leading market share. Notably, Fantasy Solution has participated in the formulation of many national IT standards and specifications.

Fantasy solution has the world's leading product engineering capabilities, ranging from consultation, design, R&D, and integration to testing of embedded software, in the fields of automotive electronics, smart devices, digital home products, and IT products. The software provided by fantasy solution runs in a number of globally renowned brands.

Particularly offering the services that include application development & maintenance, ERP implementation & consulting, testing, performance engineering, software localization & globalization, IT infrastructures, BPO, IT education & training, etc. Sticking to its business philosophy and brand commitment of “Beyond Technology”, fantasy solution provides innovative information technologies to drive the sustainable development of society, as well as becoming a company that is well recognized and respected by employees, shareholders, customers, and society.

1.2 Objectives of the Internship

The main objective of this internship was to gain practical experience in **Data Science and Machine Learning** by working on real-world data analysis and predictive applications The internship aimed to improve my knowledge of Python, data preprocessing, exploratory data analysis, data visualization, Natural Language Processing (NLP), and machine learning algorithms. Another objective was to understand the complete data science workflow, including data collection, cleaning, feature extraction, model development, evaluation, and interpretation of results.

1.3 Introduction of the Internship

The internship provided an opportunity to gain practical knowledge and hands-on experience in the field of Data Science and Machine Learning by applying theoretical concepts to real-world data. The internship focused on understanding the complete data science lifecycle, including data collection, data preprocessing, exploratory data analysis, data visualization, feature extraction, machine learning model development, and performance evaluation. As part of the internship, Natural Language Processing (NLP) techniques were studied and applied to develop a Sentiment Analysis system for customer reviews. The internship helped

in understanding how data-driven techniques can be used to extract meaningful insights from textual data and support effective decision-making.

DAILY WORK REPORT

Day 1

Learned the fundamentals of statistics, probability, and data visualization using Python and R.

Day 2

Studied data preprocessing, including missing-value handling, outlier detection, and feature scaling.

Day 3

Explored data analysis and visualization using Matplotlib and Seaborn to identify patterns and trends.

Day 4

Learned the basics of supervised and unsupervised machine learning and studied common algorithms.

Day 5

Studied model evaluation using accuracy, precision, recall, train-test splitting, and cross-validation.

Day 6

Learned feature engineering, including encoding, feature scaling, and dimensionality reduction.

Day 7

Explored model selection and optimization, including hyperparameter tuning and regularization.

Day 8

Studied classification algorithms such as Logistic Regression, SVM, and Random Forest.

Day 9

Learned regression techniques including Linear Regression, Polynomial Regression, and Gradient Boosting.

Day 10

Studied clustering algorithms such as K-Means, Hierarchical Clustering, and DBSCAN.

Day 11

Explored dimensionality reduction techniques including PCA and t-SNE.

Day 12

Learned Natural Language Processing (NLP), including text preprocessing, sentiment analysis, and text classification.

Day 13

Studied time series analysis, forecasting, ARIMA, and seasonal decomposition.

Day 14

Explored recommendation systems, including collaborative filtering and content-based recommendation.

Day 15

Learned about ensemble learning, including bagging, boosting, and stacking.

Day 16

Studied the fundamentals of deep learning, including ANN, CNN, and RNN.

Day 17

Explored neural network architecture, activation functions, and gradient descent.

Day 18

Learned about transfer learning and the use of pre-trained deep learning models.

Day 19

Studied computer vision techniques such as image classification, object detection, and segmentation.

Day 20

Explored NLP with deep learning, including word embeddings and RNN-based models.

Day 21

Learned interactive data visualization using tools such as Tableau and Plotly.

Day 22

Studied Big Data technologies, including Hadoop, Spark, and MapReduce.

Day 23

Learned SQL and database fundamentals, including queries, tables, and data manipulation.

Day 24

Explored real-time data processing using technologies such as Kafka and Spark Streaming.

Day 25

Studied machine learning model deployment using Flask, FastAPI, Docker, and cloud platforms.

Day 26

Learned about model monitoring, performance tracking, and retraining to handle changing data patterns.

Day 27

Reviewed the concepts learned during the internship and prepared a summary presentation of the work completed.

Day 28

Studied data pipelines and workflow automation using tools such as Apache Airflow and cron jobs.

Day 29

Explored cloud computing and storage using platforms such as AWS and Google Cloud.

Day 30

Revised advanced machine learning and ensemble learning techniques, including bagging, boosting, and stacking.

CHAPTER 2: LITERATURE REVIEW / SYSTEM STUDY

2.1 Evaluation of Existing Frameworks

Sentiment analysis is an important application of Natural Language Processing (NLP) and Machine Learning that is widely used to identify opinions and emotions expressed in textual data. Existing sentiment analysis systems use different methodologies, ranging from manual analysis and rule-based techniques to machine learning and deep learning approaches.

In traditional approaches, customer reviews are manually examined and categorized based on the opinions expressed by users. Although manual analysis can provide accurate results for a small number of reviews, it becomes time-consuming and inefficient when dealing with a large volume of textual data. Rule-based sentiment analysis systems use predefined dictionaries containing positive and negative words to determine the sentiment of a review. However, these systems may have difficulty understanding context, negation, slang, and complex sentences.

Machine learning-based approaches have improved sentiment classification by learning patterns from labeled datasets. Commonly used algorithms include Naive Bayes, Logistic Regression, Support Vector Machine (SVM), and Decision Tree classifiers. These approaches generally involve text preprocessing followed by feature extraction techniques such as Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF).

Modern sentiment analysis systems also use deep learning and transformer-based models such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and BERT. These models can understand contextual relationships in text more effectively but generally require greater computational resources, larger datasets, and more complex implementation.

For this project, a traditional machine learning workflow using Python, Pandas, NumPy, NLTK, Scikit-learn, and TF-IDF was considered suitable because it provides an effective balance between implementation complexity, computational requirements, interpretability, and classification performance.

2.2 Problem Statement Identification

With the rapid growth of e-commerce platforms, social media, and online services, a large amount of customer feedback is generated in the form of textual reviews. These reviews contain valuable information about customer satisfaction, product quality, service experience, and user expectations. However, manually analyzing a large number of reviews is difficult, time-consuming, and inefficient.

Traditional manual and rule-based approaches have several limitations. Manual analysis requires considerable human effort and may result in inconsistent classification when processing large datasets. Rule-based systems depend heavily on predefined sentiment dictionaries and may fail to correctly interpret contextual information, negation, abbreviations, or variations in language.

Another challenge is converting unstructured textual reviews into a format that machine learning algorithms can process effectively. Raw text may contain unnecessary characters, punctuation, stop words, duplicate information, and inconsistent formatting. Therefore, an efficient system is required to preprocess textual data, extract meaningful features, and automatically classify the sentiment of customer reviews.

The identified problem is to develop a **reliable and efficient sentiment analysis system** that can process customer reviews and classify them into **positive, negative, and neutral categories** while providing measurable performance through standard machine learning evaluation metrics. The system should reduce manual effort, handle large volumes of textual data, and provide useful insights from customer feedback.

2.3 Proposed Solution / Methodology

The proposed system is a **Machine Learning-based Sentiment Analysis System** designed to automatically analyze customer reviews and classify them into positive, negative, and neutral sentiments. The system follows a structured Data Science workflow consisting of data collection, preprocessing, exploratory data analysis, feature extraction, model training, prediction, and evaluation.

Initially, a suitable dataset containing customer reviews and their corresponding sentiment labels is collected. The collected data is examined to identify missing values, duplicate records, irrelevant information, and inconsistencies. The textual data is then preprocessed using NLP techniques such as **lowercase conversion, removal of punctuation and special characters, tokenization, stop-word removal, and text normalization**.

After preprocessing, the textual reviews are converted into numerical feature representations using **TF-IDF (Term Frequency-Inverse Document Frequency)**. The processed dataset is divided into training and testing sets. Multiple machine learning algorithms, such as **Naive Bayes, Logistic Regression, and Support Vector Machine (SVM)**, are trained using the training data.

The trained models are evaluated using **accuracy, precision, recall, F1-score, and confusion matrix**. The performance of the models is compared, and the most suitable model is selected for sentiment prediction. The final system accepts customer review text as input and predicts whether the review expresses a **positive, negative, or neutral sentiment**.

The proposed methodology provides an automated and scalable approach for analyzing customer feedback. It reduces the effort required for manual review analysis and enables organizations to extract meaningful insights from large volumes of textual customer data.

CHAPTER 3: SYSTEM REQUIREMENTS & ARCHITECTURE

3.1 Hardware & Software Prerequisites

The development of the Sentiment Analysis System requires a standard computer system capable of handling data preprocessing, exploratory data analysis, machine learning model training, and evaluation. Since the proposed system uses traditional machine learning algorithms and NLP techniques, a dedicated GPU is not mandatory. The following hardware and software requirements were considered for the development and execution of the project.

3.1.1 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 / AMD Ryzen 3
RAM	4 GB
Storage	10 GB available space
GPU	Not Required
Internet	Required for dataset/library installation
Keyboard & Mouse	Standard

The project does not require a dedicated graphics processing unit because the selected machine learning algorithms, including Naive Bayes, Logistic Regression, and Support Vector Machine, can be trained efficiently using CPU resources for a moderate-sized dataset.

3.1.2 Software Requirements

Software / Tool	Purpose
Operating System	Windows / Linux
Programming Language	Python
Development Environment	Jupyter Notebook / VS Code
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
NLP Processing	NLTK
Machine Learning	Scikit-learn
Feature Extraction	TF-IDF Vectorizer
Dataset Format	CSV
Version Control	Git (Optional)

3.1.3 Python Libraries

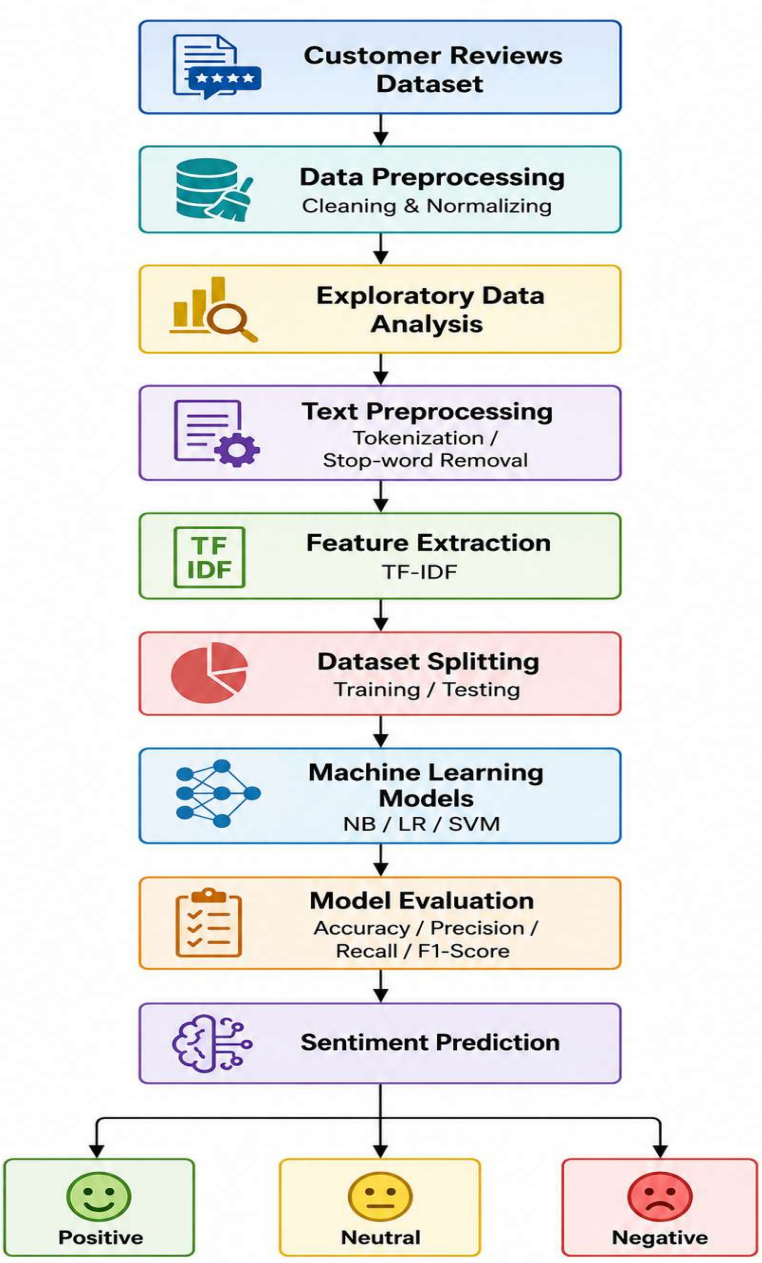
The following Python libraries are used in the implementation:

- **Pandas** – Data loading, manipulation, and analysis.
- **NumPy** – Numerical and mathematical operations.
- **Matplotlib** – Data visualization and graphical representation.
- **Seaborn** – Statistical visualization.
- **NLTK** – Natural Language Processing and text preprocessing.
- **Scikit-learn** – Feature extraction, machine learning algorithms, and evaluation metrics.
- **Regular Expressions (re)** – Identification and removal of unwanted text .

3.2 Architectural Design

The architecture of the proposed Sentiment Analysis System represents the complete flow of data from the input dataset to the final sentiment prediction. The system follows a sequential Data Science and Machine Learning workflow.

3.2.1 System Architecture



3.3 Modular Breakdown

The proposed Sentiment Analysis System is divided into several functional modules. Each module performs a specific stage of the Data Science workflow and contributes to the final sentiment classification process.

Module 1: Data Collection

This module is responsible for obtaining the customer review dataset required for model development. The dataset contains textual reviews along with their corresponding sentiment labels. The collected data is stored in a structured format such as CSV for further processing.

Functions:

- Import dataset
- Identify relevant columns
- Inspect dataset structure
- Check dataset size and class distribution

Module 2: Data Preprocessing

This module prepares the raw dataset for analysis and machine learning. Unnecessary and inconsistent information is removed from the dataset.

Functions:

- Handle missing values
- Remove duplicate records
- Remove unnecessary characters
- Normalize text
- Handle inconsistent data

Module 3: Natural Language Processing

This module performs text-specific preprocessing to convert raw customer reviews into cleaner textual data.

Functions:

- Convert text to lowercase
- Remove punctuation and special characters
- Tokenization
- Stop-word removal
- Text normalization
- Stemming or lemmatization where required

Module 4: Exploratory Data Analysis

This module analyzes the processed dataset to identify patterns and relationships within the data.

Functions:

- Analyze sentiment distribution
- Calculate review statistics
- Identify frequently occurring words
- Generate word-frequency visualizations
- Create charts and graphs

Module 5: Feature Extraction

The cleaned textual data cannot be directly processed by most traditional machine learning algorithms. Therefore, this module converts text into numerical representations using **TF-IDF (Term Frequency-Inverse Document Frequency)**.

Functions:

- Generate vocabulary
- Calculate TF-IDF values
- Create feature vectors
- Prepare numerical input for machine learning models

Module 6: Machine Learning Model Development

This module trains multiple machine learning algorithms using the extracted features.

Models considered:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)

The models learn patterns from the training dataset and use those patterns to classify unseen reviews.

Module 7: Model Evaluation

This module evaluates and compares the performance of the trained models.

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The model with the most suitable performance is selected for the final prediction system.

Module 8: Sentiment Prediction

The final module uses the selected trained model to classify new customer reviews.

Input:

"The product quality is excellent and I am very satisfied."

Processing:

Text preprocessing → TF-IDF transformation → trained model

Output:

Positive Sentiment

The same process is applied to negative and neutral reviews to provide automated sentiment classification.

CHAPTER 4: DEVELOPMENT AND RESULT ANALYSIS

4.1 System Implementation and Development

The Sentiment Analysis System was developed using a structured Data Science and Machine Learning workflow. The implementation was carried out using **Python** along with libraries such as **Pandas, NumPy, NLTK, Matplotlib, Seaborn, and Scikit-learn**. The development process involved data collection, data preprocessing, exploratory data analysis, feature extraction, machine learning model training, evaluation, and sentiment prediction.

4.1.1 Dataset Loading and Data Collection

The first stage of implementation involved collecting a suitable dataset containing customer reviews and their corresponding sentiment labels. The dataset was stored in CSV format and loaded into the Python environment using the Pandas library.

The dataset was initially examined to determine the number of records, available attributes, data types, missing values, duplicate entries, and distribution of sentiment classes. This initial analysis helped identify the quality and structure of the available data before applying further processing techniques.

4.1.2 Data Preprocessing

Raw customer reviews may contain unnecessary characters, punctuation, numbers, URLs, extra spaces, and other elements that can affect the performance of a machine learning model. Therefore, preprocessing was performed to convert the raw textual data into a consistent format.

The following preprocessing operations were performed:

1. Removal of missing and duplicate records.
2. Conversion of text into lowercase.
3. Removal of punctuation and special characters.
4. Removal of unnecessary spaces.
5. Tokenization of text into individual words.
6. Removal of common stop words.
7. Text normalization.
8. Stemming or lemmatization where required.

The preprocessing stage improves the quality of the textual data and prepares it for feature extraction.

4.1.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics and distribution of the dataset. Different statistical and graphical techniques were used to identify patterns in customer reviews.

The analysis included:

- Distribution of positive, negative, and neutral reviews.
- Frequency of commonly occurring words.
- Review length analysis.
- Identification of frequently used terms.
- Graphical representation of sentiment classes.

Libraries such as **Matplotlib and Seaborn** were used to generate charts and visualizations.

4.1.4 Feature Extraction Using TF-IDF

Machine learning algorithms require numerical input, whereas customer reviews are represented as textual data. Therefore, the pre-processed text was converted into numerical feature vectors using **Term Frequency-Inverse Document Frequency (TF-IDF)**.

TF-IDF assigns a numerical importance score to words based on their frequency within a particular review and their frequency across the complete collection of reviews. Words that provide useful information for distinguishing between different sentiment classes receive higher importance.

The resulting TF-IDF matrix was used as the input for training the machine learning models.

4.1.5 Dataset Splitting

The processed dataset was divided into two major subsets:

- **Training Dataset** – Used to train the machine learning models.
- **Testing Dataset** – Used to evaluate the performance of the trained models on previously unseen data.

A suitable train-test split was applied to ensure that the model evaluation represented its ability to generalize to new customer reviews.

4.1.6 Machine Learning Model Development

Multiple supervised machine learning algorithms were implemented to identify the most suitable model for sentiment classification.

The following algorithms were considered:

Naive Bayes

Naive Bayes is a probabilistic classification algorithm that is commonly used for text classification. It calculates the probability of a review belonging to a particular sentiment class based on the features extracted from the text.

Logistic Regression

Logistic Regression is a classification algorithm that estimates the probability of a data instance belonging to different classes. It is widely used for text classification because of its simplicity, efficiency, and good performance with high-dimensional feature spaces.

Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm that identifies an optimal decision boundary between different classes. It is particularly effective for text classification problems involving a large number of features.

The models were trained using the TF-IDF feature vectors and the corresponding sentiment labels.

4.1.7 Model Evaluation and Selection

After training, each model was evaluated using the testing dataset. The predictions generated by the models were compared with the actual sentiment labels.

The following evaluation metrics were used:

- Accuracy
- Precision

- Recall
- F1-score
- Confusion Matrix

The performance of the models was compared, and the model providing the most suitable overall performance was selected as the final sentiment classification model.

4.1.8 Sentiment Prediction

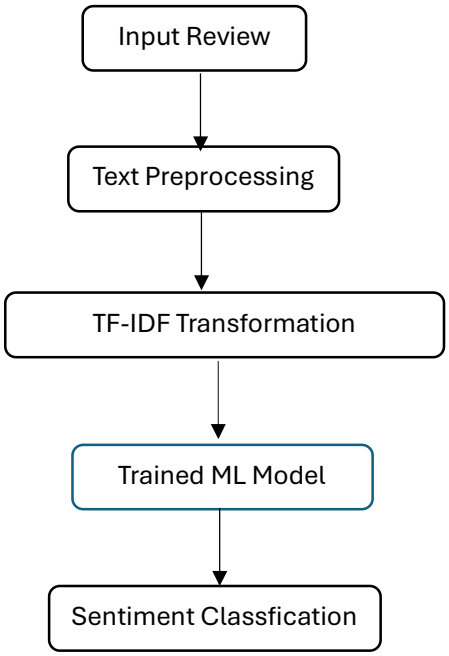
After selecting the final model, a prediction pipeline was developed to classify new customer reviews.

For example:

Input:

"The product quality is excellent and I am very satisfied."

Processing:



Output:

Positive Sentiment

The same process can classify reviews as negative or neutral based on the patterns learned during model training.

4.2 Result Analysis and Testing Parameters

The performance of the developed Sentiment Analysis System was analyzed using standard machine learning evaluation metrics. The testing process was performed using unseen test data to determine how effectively the trained models classified customer reviews.

4.2.1 Evaluation Parameters

Accuracy

Accuracy represents the percentage of correctly classified reviews among all the reviews tested.

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Precision

Precision represents the proportion of correctly predicted positive classifications among all instances predicted as positive.

Formula:

Precision = TP / (TP + FP)

A higher precision indicates fewer false-positive predictions.

Recall

Recall represents the proportion of correctly identified actual positive instances.

Formula:

Recall = TP / (TP + FN)

A higher recall indicates that the model is able to identify a greater number of relevant instances.

F1-Score

F1-score is the harmonic mean of precision and recall.

Formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

F1-score is particularly useful when evaluating classification performance across different sentiment classes.

4.2.2 Confusion Matrix

A confusion matrix was used to analyse the classification performance of the model for each sentiment category. Since the proposed system contains three sentiment classes, the confusion matrix represents the relationship between:

- **Positive**
- **Negative**
- **Neutral**

Correct predictions appear along the diagonal of the matrix, while incorrect classifications appear in the other cells.

4.2.3 Model Performance Comparison

The performance of the implemented machine learning algorithms will be compared using the following parameters:

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	To be calculated	To be calculated	To be calculated	To be calculated
Logistic Regression	To be calculated	To be calculated	To be calculated	To be calculated
Support Vector Machine	To be calculated	To be calculated	To be calculated	To be calculated

The comparison helps identify which machine learning algorithm provides the best performance for the selected customer review dataset.

4.2.4 Functional Testing

Functional testing was performed to verify whether each major component of the system operates according to the expected requirements.

Test Case	Input / Condition	Expected Result
TC01	Load valid dataset	Dataset loaded successfully
TC02	Dataset contains missing values	Missing records identified/handled

Test Case	Input / Condition	Expected Result
TC03	Raw review provided	Text preprocessing performed
TC04	Pre-processed review	TF-IDF features generated
TC05	Training dataset provided	Model trained successfully
TC06	Test dataset provided	Sentiment prediction generated
TC07	Positive review entered	Positive sentiment predicted
TC08	Negative review entered	Negative sentiment predicted
TC09	Neutral review entered	Neutral sentiment predicted
TC10	Model evaluation executed	Evaluation metrics generated

4.2.5 Sample Prediction Testing

The developed system can be tested using sample customer reviews to verify the prediction pipeline.

Sample Review	Expected Sentiment	Predicted Sentiment
"The product is excellent and worth the price."	Positive	To be generated
"The product quality is very poor."	Negative	To be generated
"The product is okay for its price."	Neutral	To be generated

The predicted results will be obtained after executing the trained model. The actual outputs should be recorded in the final report rather than manually assigning prediction values.

4.2.6 Code implementation

```
# SENTIMENT ANALYSIS OF CUSTOMER REVIEWS

# Using NLP, TF-IDF and Logistic Regression


import pandas as pd

import re

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("customer_reviews.csv")

print("Dataset loaded successfully")

print(data.head())

def preprocess_text(text):

    text = str(text).lower()

    text = re.sub(r'http\S+|www\S+', '', text)

    text = re.sub(r'[^a-zA-Z\s]', '', text)

    text = re.sub(r'\s+', ' ', text).strip()
```

```
    return text

data["clean_review"] = data["review"].apply(preprocess_text)

print("\nPreprocessed Reviews:")

print(data[["review", "clean_review"]].head())

X = data["clean_review"]

y = data["sentiment"]

X_train, X_test, y_train, y_test = train_test_split(X, y
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining samples:", len(X_train))

print("Testing samples :", len(X_test))

tfidf = TfidfVectorizer(
    max_features=5000,
    stop_words="english"
)

X_train_tfidf = tfidf.fit_transform(X_train)

X_test_tfidf = tfidf.transform(X_test)

print("\nTF-IDF feature extraction completed.")

print("Number of features:", len(tfidf.get_feature_names_out()))

model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train_tfidf, y_train)

print("\nModel training completed successfully.")

y_pred = model.predict(X_test_tfidf)

accuracy = accuracy_score(y_test, y_pred)

print("\n====================================")

print("       MODEL PERFORMANCE")

print("====================================")

print("Accuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")

print(classification_report(y_test, y_pred))

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred))

new_reviews = [

    "The product is excellent and I am very happy with it.",

    "The quality is extremely poor and disappointing.",

    "The product is okay and works as expected."

]

new_reviews_clean = [

    preprocess_text(review)

    for review in new_reviews

]
```

```
new_reviews_tfidf = tfidf.transform(new_reviews_clean)

predictions = model.predict(new_reviews_tfidf)

print("\n====================================")

print("       SENTIMENT PREDICTIONS")

print("====================================")

for review, sentiment in zip(new_reviews, predictions):

    print("\nReview:", review)

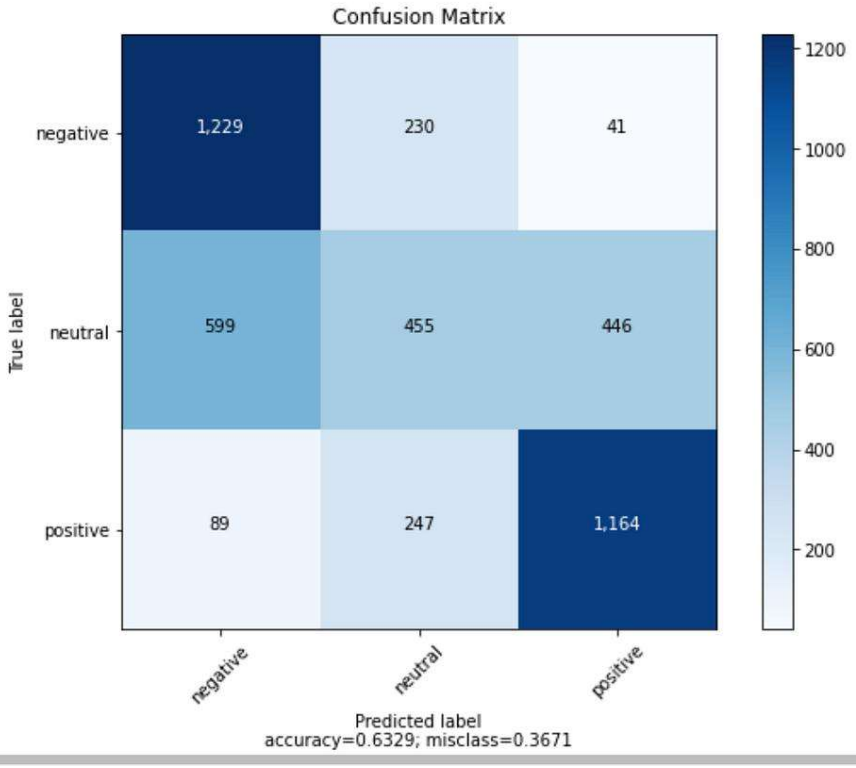
    print("Predicted Sentiment:", sentiment)
```

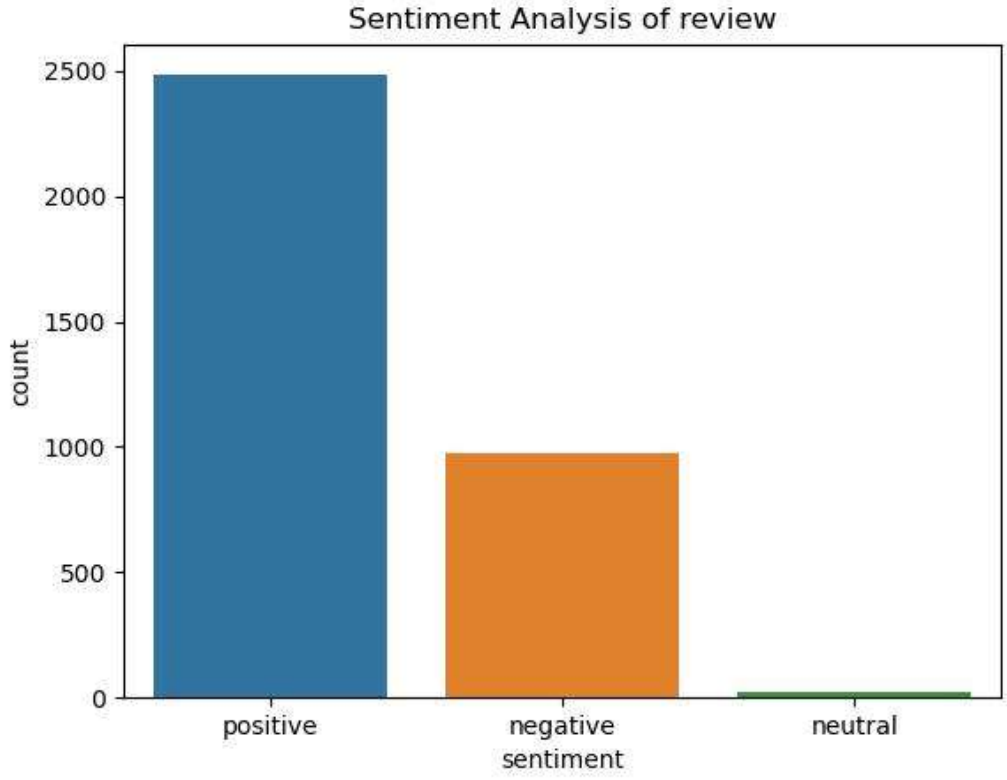
4.2.7 Sample Dataset

Review	Sentiment
The product is excellent and very useful	Positive
I am extremely happy with the purchase	Positive
Amazing quality and fast delivery	Positive
The product is terrible	Negative
Very poor quality and disappointing	Negative
I regret buying this product	Negative
The product is okay	Neutral
It works as expected	Neutral
Average product for the price	Neutral

4.2.8 Sample Output

```
[87]: plot_confusion_matrix(cm           = cm,
                            normalize    = False,
                            target_names = ["negative", "neutral", "positive"],
                            title        = "Confusion Matrix")
```






```
[3]
test_dataset = [
    "I love using Hugging Face for NLP.",
    "This is confusing and frustrating.",
    "The documentation is detailed but overwhelming.",
    "Seriously, I hate how overwhelming this is!",
    "But, AI helps me a lot in my daily tasks.",
]

✓ 0.0s
```

```
[4]
result = pipe(test_dataset)

result

✓ 8.9s
```

```
[{'label': 'positive', 'score': 0.978042483329773},
 {'label': 'negative', 'score': 0.8942117094993591},
 {'label': 'negative', 'score': 0.5077610015869141},
 {'label': 'negative', 'score': 0.9151564836502075},
 {'label': 'positive', 'score': 0.927707850933075}]
```

4.2.9 Result Analysis

The final results will be analysed by comparing the performance of the different machine learning models. The model with the highest overall performance based on accuracy, precision, recall, and F1-score will be considered the most suitable model for the proposed sentiment analysis system.

The confusion matrix will further help identify the types of classification errors made by the selected model, such as incorrectly classifying a neutral review as positive or a negative review as neutral.

The results demonstrate the effectiveness of combining **Natural Language Processing, TF-IDF feature extraction, and supervised machine learning** for automated sentiment classification of customer reviews.

4.3 Future Enhancements

Although the proposed Sentiment Analysis System provides an effective approach for classifying customer reviews, several enhancements can be introduced to improve its accuracy, scalability, and practical usability in the future.

1. **Deep Learning Models:**
   Future versions can incorporate advanced deep learning models such as LSTM, GRU, and Transformer-based architectures to improve sentiment classification performance.
2. **BERT and Transformer Models:**
   Models such as BERT can be integrated to understand the contextual meaning of words and sentences more effectively, particularly when dealing with complex customer opinions.
3. **Multilingual Sentiment Analysis:**
   The system can be enhanced to support multiple languages such as Tamil, Hindi, and other regional languages, making it useful for a wider range of users.

4. **Aspect-Based Sentiment Analysis:**
   Instead of assigning a single sentiment to an entire review, the system can identify sentiments associated with individual aspects such as **price, quality, delivery, customer service, and product features**.
5. **Real-Time Sentiment Analysis:**
   The system can be integrated with social media platforms, e-commerce websites, or customer-feedback systems to perform sentiment analysis on newly received reviews in real time.

CHAPTER 5 : CONCLUSION

The **Sentiment Analysis of Customer Reviews Using Natural Language Processing and Machine Learning** project provided practical experience in applying Data Science concepts to a real-world problem. The project demonstrated how unstructured textual customer reviews can be processed, analyzed, and transformed into useful information using Natural Language Processing and Machine Learning techniques.

The developed workflow included **data collection, data preprocessing, exploratory data analysis, text preprocessing, TF-IDF feature extraction, machine learning model development, and performance evaluation**. Different classification algorithms can be evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix to identify the most suitable model for the given dataset.

Through this project, practical knowledge was gained in **Python, Pandas, NumPy, NLTK, Matplotlib, Seaborn, Scikit-learn, NLP, and Machine Learning**. The internship also provided an understanding of the complete Data Science lifecycle, from handling raw data to developing and evaluating a machine learning solution.

Overall, the project demonstrates that sentiment analysis can reduce the effort involved in manually analyzing large volumes of customer feedback and can help organizations understand customer opinions more efficiently. The knowledge and practical experience gained during the internship provide a strong foundation for developing more advanced Data Science and Machine Learning applications in the future.

5.1 REFERENCES

1.Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze, **Introduction to Information Retrieval**, Cambridge University Press, 2008.

2.Steven Bird, Ewan Klein, and Edward Loper, **Natural Language Processing with Python**, O'Reilly Media, 2009.

3.Aurélien Géron, **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow**, 3rd Edition, O'Reilly Media, 2022.

4.Pang, B., Lee, L., and Vaithyanathan, S., **"Thumbs Up? Sentiment Classification using Machine Learning Techniques,"** Proceedings of the Conference on Empirical Methods in Natural Language Processing, 2002.

5.Liu, B., **Sentiment Analysis and Opinion Mining**, Morgan & Claypool Publishers, 2012.

6. Pedregosa, F., Varoquaux, G., Gramfort, A., et al., **"Scikit-learn: Machine Learning in Python,"** Journal of Machine Learning Research, Vol. 12, pp. 2825–2830, 2011.

7.Python Documentation

8. Pandas Documentation

9. NLTK Documentation

ANNEXURE

Internship (June – July 2026)

1.Name of the Industry with address:

Fantasy Solution, No 16, Samnath Plazza, Third Floor, Near BG Naidu Sweets, Melapudur, Trichy-1.

2. Name of the Industrial Expert (Joint Supervisor): Mohamed Ibrahim

3. Qualification of the Industrial Expert (Joint Supervisor): MCA

4. Designation of the Industrial Expert (Joint Supervisor): Software Developer

5. Title of the Project Work:

Sentiment Analysis of Customer Reviews Using Natural Language Processing and Machine Learning

6. Duration of the Internship: 1st June 2026 – 30th June 2026

7. Details of the student:

Sl. No.	Name of the student	Registration No.	Department
01	VIGNESHWARAN L	810023205062	IT

Terms and conditions

- The student is permitted to undergo a semester long internship in your organization
- The student shall be instructed to meet the Institute supervisor periodically and to attend the review committee meetings for evaluating the progress.
- The final report of the internship duly signed by the industrial expert/supervisor will be submitted to college.
- Attendance Certificate in the prescribed format signed by the competent authority of the industry will be produced
- Internship Certificate in the prescribed format signed by the competent authority of the industry will be produced

Seal of the Organization

Signature and Designation of the Officer